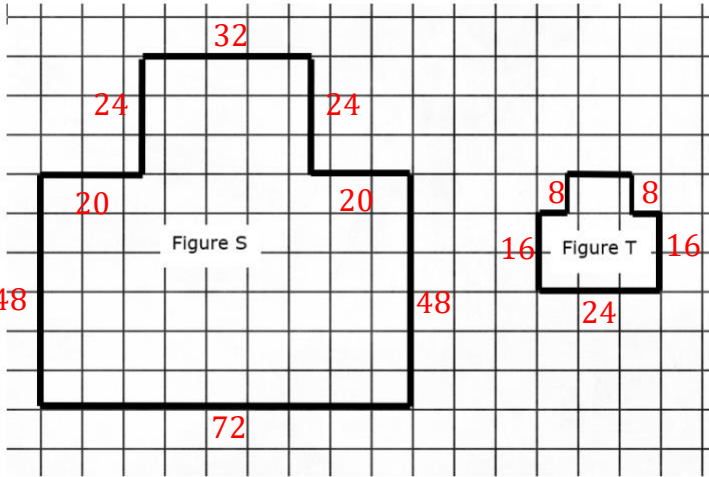


Figure S was scaled down and drawn as Figure T. Each square represents a width of 8 units. Use this information to complete problems 1-5.



1. Label 3 pairs of corresponding sides from the original to the scaled down drawing lines from Figure S to Figure T.
- Pairs use may vary. The sides not labeled in Figure T due to space are $10\frac{2}{3}$ units (top) and $6\frac{2}{3}$ (two horizontal sides that are less than 1 unit each.) Students can draw arrows between any pairs shown in the table.

Side length on Figure S	Side Length on Figure T
72	24
48	16
20	$6\frac{2}{3}$
24	8
32	$10\frac{2}{3}$

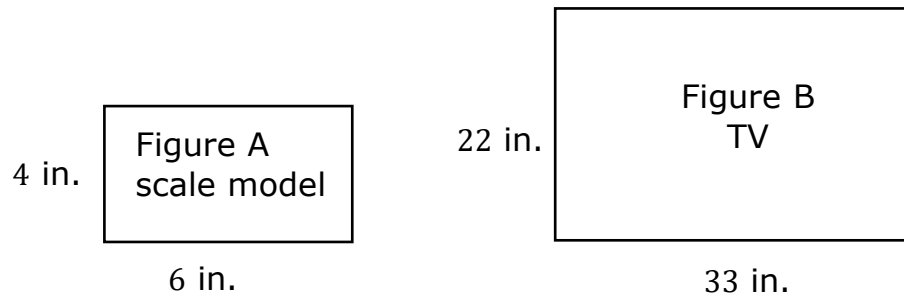
2. What is the scale factor from Figure T to Figure S?
 $\text{Figure T} \times 3 = \text{Figure S}$

3. What scale factor takes Figure S to Figure T?
 $\text{Figure S} \times \frac{1}{3} = \text{Figure T}$

4. What is the estimated perimeter of Figure S?
 $72 + 2(48) + 2(20) + 2(24) + 32 = 288 \text{ units}$

5. What is the ratio of the perimeter Figure T to the perimeter of Figure S?
- $\frac{\text{Perimeter T}}{\text{Perimeter S}} = \frac{96}{288} = \frac{1}{3}$

Shannon draws a scale model of her TV screen, Figure A, with a width of 6 in. and height of 4 in. Her actual TV, Figure B, has a width of 33 in. and height of 22 in. Use this information to complete problems 6-10.



6. What is the scale factor from Figure A to Figure B?

$$\text{Figure A} \times 5.5 = \text{Figure B}$$

7. What is the perimeter of Shannon's scale model?

$$4(2) + 6(2) = 20 \text{ in.}$$

8. What is the perimeter of Shannon's TV?

$$2(22) + 2(33) = 110 \text{ in.}$$

9. What is the ratio of the perimeter of Shannon's scale model to her actual TV?

$$\frac{20}{110} = \frac{2}{11}$$

10. How does the scale factor from Shannon's actual TV to the scale model compare to the scale factor from her scale model to her actual TV? Explain.

$$\frac{2}{11} \text{ is the reciprocal of } \frac{11}{2} (5.5).$$